

Chapter 3 – Preparing the Virtual Machine

Objective

In this chapter, we will create an Ubuntu Server virtual machine using Oracle VirtualBox.

By the end of this chapter, you will have a VM ready for installing Ubuntu Server.

3.1 Why Use Virtual Machines?

A Virtual Machine (VM) allows multiple operating systems to run on a single physical computer.

Benefits include:

- Safe testing environment
- No additional hardware required
- Snapshot support
- Isolation from the host operating system
- Easy recovery from mistakes

For cybersecurity, VMs make it possible to build complete labs using one computer.

3.2 Install Oracle VirtualBox

Download VirtualBox from the official website:

<https://www.virtualbox.org/>

Install both:

- Oracle VirtualBox
- Oracle VirtualBox Extension Pack

The Extension Pack adds support for:

- USB 2.0 / 3.0
 - Remote Desktop (VRDP)
 - Disk encryption
 - NVMe support
 - PXE boot for Intel NICs
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3.3 Create a New Virtual Machine

Open VirtualBox.

Select:

Machine → **New**

Enter the following settings:

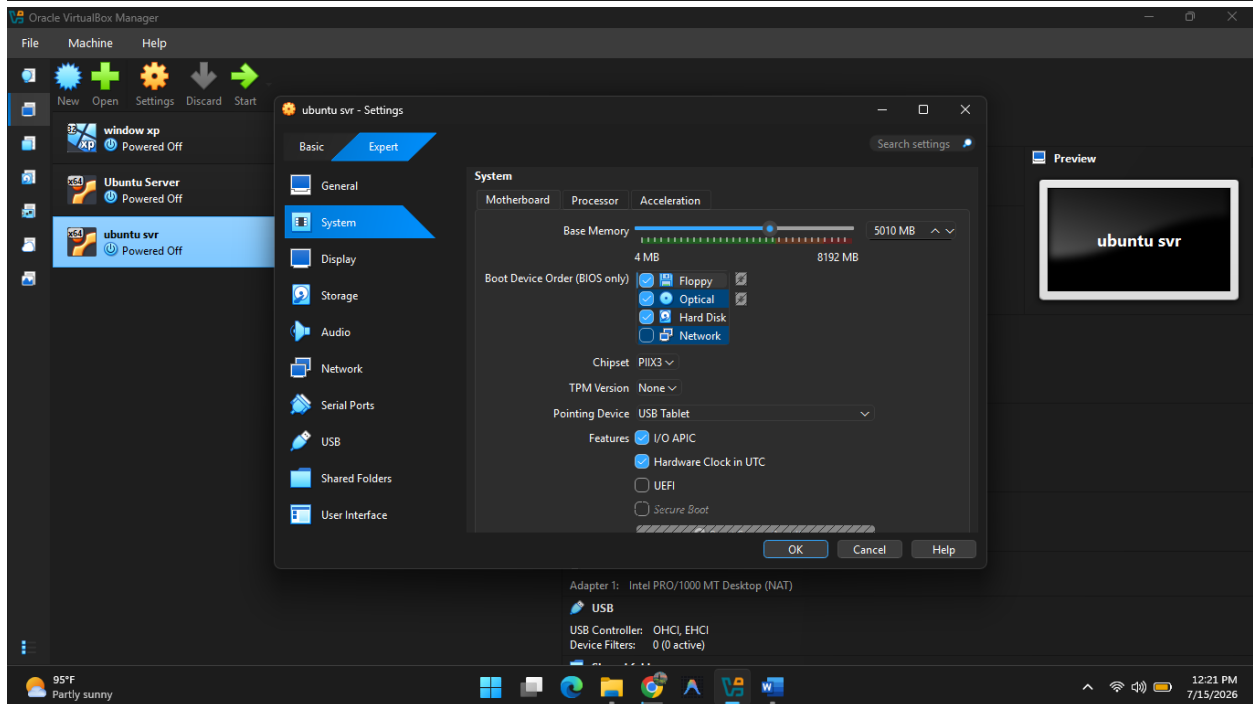
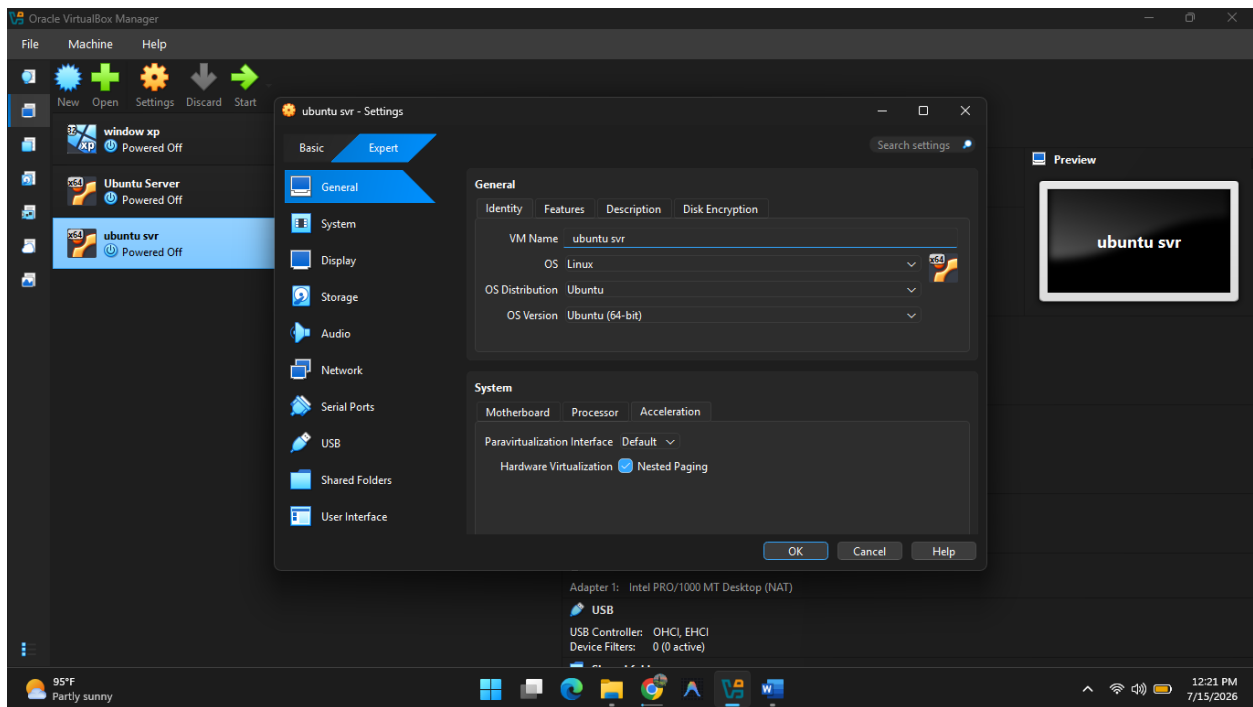
Setting	Value
Name	Ubuntu-Server
Type	Linux
Version	Ubuntu (64-bit)

3.4 Allocate Memory

Recommended:

RAM	Value
Minimum	4 GB
Recommended	6–8 GB

Ubuntu Server itself uses little memory, but additional services such as Wazuh require more resources.



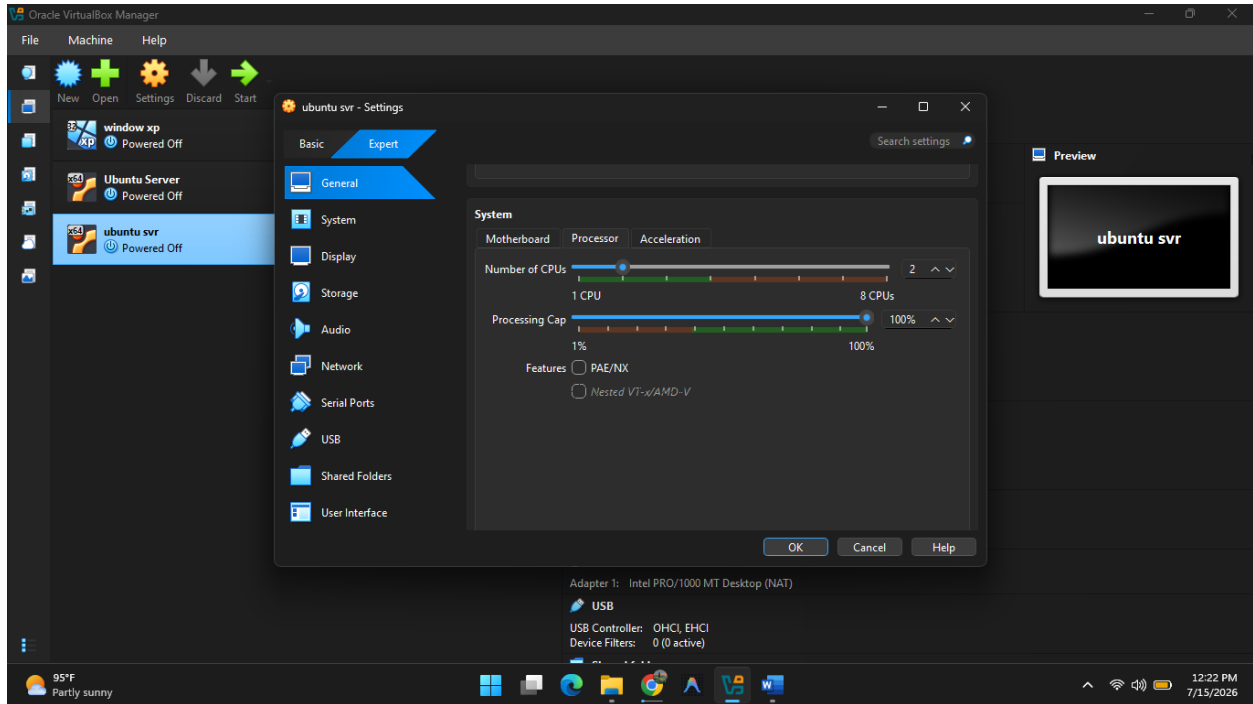
3.5 Configure the Processor

Recommended:

CPUs	Value
Minimum	2

Recommended	2–4
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Using multiple CPU cores improves responsiveness when running server applications.



3.6 Create a Virtual Hard Disk

Recommended settings:

Option	Value
Disk Type	VDI
Storage	Dynamically Allocated
Size	80 GB

A dynamically allocated disk grows as data is written instead of occupying the full size immediately.

3.7 Attach the Ubuntu ISO

Open:

Settings → Storage

Select the empty optical drive and attach the Ubuntu Server 22.04.5 LTS ISO downloaded earlier.

This allows the VM to boot from the installation media.

3.8 Configure the Network

For now, configure:

Adapter Mode

Adapter 1 NAT

A Host-Only Adapter will be added after the operating system is installed to support communication with other virtual machines in the lab.

3.9 Review the Configuration

Before starting the VM, verify:

- Ubuntu Server ISO attached
- 2–4 CPU cores assigned
- 4–8 GB RAM allocated
- 80 GB dynamic virtual disk
- NAT networking enabled

If everything looks correct, start the virtual machine.

Best Practices

- Use dynamic disks unless fixed-size storage is specifically required.
- Keep VM names descriptive.

- Store VMs on an SSD if available.
 - Enable virtualization (Intel VT-x or AMD-V) in the BIOS if VirtualBox reports hardware virtualization errors.
 - Take a snapshot after the operating system installation completes successfully.
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Chapter Summary

In this chapter, you learned how to:

- Create a VirtualBox virtual machine.
 - Allocate CPU, RAM, and storage.
 - Attach the Ubuntu Server ISO.
 - Configure initial networking.
 - Prepare the VM for installation.
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One-Line Summary

A properly configured virtual machine provides a reliable and isolated environment for installing and managing Ubuntu Server in a cybersecurity home lab.
